USN [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] **[22CSBS233]**

# SECOND SEMESTER B.Tech EXAMINATION, APRIL - 2023

## STATISTICAL METHODS WITH R-LAB

**TIME: 3 HOURS** **Maximum Marks: 100**

**Instructions:**

i. ANSWER ALL QUESTIONS

ii. Missing data may be suitably assumed.

**PART-A** **5 X 2=10**

| | | | | | |
|---|---|---|---|---|---|
| **1.** | **a.** | A population consist of five numbers 3, 5, 7, 9, 11. consider all possible samples of size 2 which can be drawn with replacement from the population. Find mean and standard deviation of the population | L2 | CO1 | **02** |
| | **b.** | Show that $r = \sqrt{b_{xy}\, b_{yx}}$ where $r$ is the correlation coefficient and $b_{xy}, b_{yx}$ are regression coefficients. | L2 | CO2 | **02** |
| | **c.** | Explain One tailed test and Two tailed test. | L1 | CO3 | **02** |
| | **d.** | Explain Run test. | L1 | CO4 | **02** |
| | **e.** | Solve 4-yearly moving average for the following data: | L3 | CO5 | **02** |

| Year | 2001-02 | 2002-03 | 2003-04 | 2004-05 | 2005-06 | 2006-07 | 2007-08 | 2008-09 |
|---|---|---|---|---|---|---|---|---|
| Production | 40 | 45 | 40 | 42 | 46 | 52 | 56 | 61 |

**PART-B** **6 X 5=30**

| | | | | | |
|---|---|---|---|---|---|
| **2.** | **a.** | Describe Non probabilistic sampling methods. | L2 | CO1 | **05** |
| | | **OR** | | | |
| | **b.** | Following are the weights of four bags of rice(in kgs) chosen at random from a lot of 100 bags: | L3 | CO1 | **05** |

| | | | |
|---|---|---|---|
| 101kg | 99kg | 98kg | 102kg |

Find the best estimator of true mean weight, variance and standard deviation of all the bags.

| | | | | | |
|---|---|---|---|---|---|
| **3.** | **a.** | Fit a parabola $y = a + bx + cx^2$ for the data | L3 | CO2 | **05** |

| $x$ | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| $y$ | 1 | 1.8 | 1.3 | 2.5 | 2.3 |

| | | | | | |
|---|---|---|---|---|---|
| | | **OR** | | | |
| | **b.** | Fit a curve of the form $y = ae^{bx}$ for the data | L3 | CO2 | **05** |

| $x$ | 0 | 2 | 4 |
|---|---|---|---|
| $y$ | 8.12 | 10 | 31.82 |

| | | | | | |
|---|---|---|---|---|---|
| **4.** | **a.** | Suppose $X$ is single observation sample from a population with probability density function $f(x) = \theta x^{\theta-1}$ for $0 < x < 1$. Find a test with the best critical region, taking significance level $\propto = 0.05$, for testing the sample $H_0: \theta = 3$ against the alternative hypothesis $H_1: \theta = 2$. | L3 | CO3 | **05** |
| | | **OR** | | | |
| | **b.** | The mean mathematics score for the population of first-year students at University is 520. The standard deviation of scores in this population is 95. An investigator believes that the mean mathematics score of first-year B.Tech course is significantly different from the mean score of the population. The mean of a sample of 36 first year B.Tech students is 548. Please test the investigator's prediction using an alpha level of 5%.($Z_c$=1.96) | L3 | CO3 | **05** |
| | | | | | |
| **5.** | **a.** | Explain any two Non parametric tests. | L2 | CO4 | **05** |
| | | **OR** | | | |
| | **b.** | Using Mann whitney test, compute $U_x$, $U_y$ & $U$ for the following samples | L3 | CO4 | **05** |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| X | 60 | 45 | 23 | 32 | | | |
| Y | 10 | 25 | 20 | 54 | 32 | 65 | 8 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| **6.** | **a.** | The net profits of a business have been given below, examine net profit for 2012. | L3 | CO5 | **05** |
| | | **OR** | | | |
| | **b.** | Write a note on ARIMA model? | L2 | CO5 | **05** |
| | | | | | |
| **7.** | **a.** | Compute 4-year moving averages centered for the following data | L3 | CO5 | **05** |

| Year | 1995 | 1996 | 1997 | 1998 | 1999 | 2000 | 2001 | 2002 |
|---|---|---|---|---|---|---|---|---|
| Production | 80 | 90 | 92 | 83 | 87 | 96 | 100 | 110 |

| | | | | | | |
|---|---|---|---|---|---|---|
| | | **OR** | | | | |
| | **b.** | Calculate the seasonal indices for the following data using the moving average method | L3 | CO5 | **05** |

| Year | I Quarterly | II Quarterly | III Quartely | IV Quartely |
|---|---|---|---|---|
| 2008 | 72 | 68 | 62 | 76 |
| 2009 | 78 | 74 | 78 | 72 |
| 2010 | 74 | 70 | 72 | 76 |
| 2011 | 76 | 74 | 74 | 72 |
| 2012 | 72 | 72 | 76 | 68 |

| | | PART-C | | | 6 X 10=60 |
|---|---|---|---|---|---|
| **8.** | **a.** | A population consist of five numbers 2,3,6,8,11. consider all possible samples of size 2 which can be drawn with replacement from population find<br>**i.** The mean and standard deviation of the population<br>**ii.** The mean and standard deviation of the sampling distribution of means | L3 | CO1 | **10** |
| | | **OR** | | | |
| | **b.** | **i.** Write a note on Probabilistic sampling?<br>**ii.** The Electric Light bulbs of a manufacturer A have mean life of 1300 hr with standard deviation of 120 hr, while for manufacturer B have mean life of 1200 hr with standard deviation of 90 hr. If a random samples of 150 bulb of each brand are tested, what is probability that Brand A bulbs have mean life which is at least 80 hr more than brand B bulbs? ($\phi(1.63) = 0.445$). | L3 | CO1 | **10** |
| | | | | | |
| **9.** | **a.** | Three varieties of coal were analyzed by four chemists and the ash-content in the varieties was found to be as under | L3 | CO2 | **10** |

| | Chemists | | | |
|---|---|---|---|---|
| Varieties | 1 | 2 | 3 | 4 |
| A | 8 | 5 | 5 | 7 |
| B | 7 | 6 | 4 | 4 |
| C | 3 | 6 | 5 | 4 |

Carry out analysis of variance.

| | | | | | |
|---|---|---|---|---|---|
| | | **OR** | | | |
| | **b.** | Fit a straight line y=a+bx to the following data. Also find production of the year 2006 | L3 | CO2 | **10** |

| Year | 1961 | 1971 | 1981 | 1991 | 2001 |
|---|---|---|---|---|---|
| Production(in tons) | 8 | 10 | 12 | 10 | 16 |

| | | | | | |
|---|---|---|---|---|---|
| | | | | | |
| **10.** | **a.** | **i.** Explain the advantages of parametric test.<br>**ii.** The mean mathematics score for the population of first-year students at X-University is 520. The standard deviation of scores in this population is 95. An investigator believes that the mean mathematics score of first-year B.Tech course is significantly different from the mean score of the population. The mean of a sample of 36 first year B.Tech students is 548. Please test the investigator's prediction using an alpha level of 5%.(Z-value:1.96) | L3 | CO3 | **10** |
| | | **OR** | | | |
| | **b.** | **i.** Describe parametric tests to test the significance of small and large | L3 | CO3 | **10** |

samples.

ii. Ten individuals are chosen at random from a population and their heights in inches are found to be 63, 63, 66, 67, 68, 69, 70, 70, 71,71. Test the hypothesis that the mean height of the universe is 66 inches. Use 5% LOS.($t_{0.05,9} = 2.262$)

**11. a.** i. Explain the difference between Parametric and Non Parametric tests — L4 CO4 **10**

ii. A random sample of 32 checking accounts at first state bank gives the following monthly balance in($)

185 210 324 150 165 134 165 195 245 164 155 320 175
146 148 152 126 159 164 158 211 215 249 168 146 164
157 231 194 182 168 154

Use the sign test at 5% level of significance and the test that the median monthly balance is at least $200.$Z_c$=-1.645

**OR**

**b.** In a study done from various streams of a college 60 students, with equal number of students drawn from each stream, the intention of the students to join the Adventure club of college noted after the interviewed are listed below. It was expected that 12 students from each class would join the Adventure Club. Examine by using K-S test to know if there is any difference among student classes with regard to their intention of joining the club. — L4 CO4 **10**

| | B.Sc | B.A. | B.Com | M.A. | M.Com |
|---|---|---|---|---|---|
| No. in each class | 5 | 9 | 11 | 16 | 19 |

($D_{0005} = \frac{1.36}{\sqrt{n}}$)

**12. a.** Compute 5-year, 7-year and 9-year moving averages for the following data: — L3 CO5 **10**

| Years | 1990 | 1991 | 1992 | 1993 | 1994 | 1995 | 1996 | 1997 | 1998 | 1999 | 2000 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Values | 2 | 4 | 6 | 8 | 10 | 12 | 14 | 16 | 18 | 20 | 22 |

**OR**

**b.** Fit a straight line trend by the method of least squares and examine the trend values. — L3 CO5 **10**

| Year | 1991 | 1962 | 1963 | 1964 | 1965 | 1966 | 1967 | 1968 |
|---|---|---|---|---|---|---|---|---|
| Value | 80 | 90 | 92 | 83 | 94 | 99 | 92 | 104 |

| | | | | | |
|---|---|---|---|---|---|
| **13.** | **a.** | Following table shows the frequency distribution of college student according to their pocket money (daily). Draw histogram and frequency polygon on the data given. | L3 | CO3 | **10** |

| Pocket Money (In Rs) | 20-29 | 30-39 | 40-49 | 50-59 | 60-69 | 70-79 | 80-89 |
|---|---|---|---|---|---|---|---|
| No. of Students | 10 | 24 | 18 | 12 | 8 | 5 | 6 |

Also explain R-commands to draw a histogram.

**OR**

| | | | | | |
|---|---|---|---|---|---|
| | **b.** | Compute mean for the following frequency Distribution using R-code: | L3 | CO1 | **10** |

| Height in Cm | 145-150 | 150-155 | 155-160 | 160-165 | 165-170 | 170-175 | 175-180 | 180-185 |
|---|---|---|---|---|---|---|---|---|
| No. of Adult men | 4 | 6 | 28 | 58 | 64 | 30 | 5 | 5 |

Also discuss R-code to find mean, Median and Mode.